University of Tehran
College of Engineering
School of Electrical and Computer Engineering

Machine Learning

Final Project – Phase 1

Name
Parsa KafshduziBukani

Student ID
810102501

January 6, 2026

Contents

Abstract

Spoken Language Identification (LID) aims to automatically determine the language of a speech signal based solely on its acoustic characteristics, without relying on semantic understanding. The performance of LID systems is highly dependent on the quality and representativeness of the underlying audio data, making careful dataset construction a critical prerequisite for effective modeling.

This phase of the final project focuses on the creation of a high-quality multilingual speech dataset extracted from audiobook recordings. Audio segments are selected with strict adherence to sentence boundaries to preserve natural prosody, rhythm, and phonetic structure. Emphasis is placed on qualitative listening-based evaluation to ensure clarity, sufficient recording quality, and linguistic representativeness of each sample. The collected dataset consists of short, uniform-duration speech segments that are suitable for subsequent feature extraction and machine learning analysis.

By actively participating in the data collection process, this phase highlights the importance of informed audio selection and human-in-the-loop validation in speech-based machine learning pipelines. The resulting dataset forms a reliable foundation for later phases of the project, including feature extraction, similarity learning, and language classification.

Introduction to Spoken Language Identification

Definition and Importance of Spoken Language Identification

Spoken Language Identification (LID) is the task of automatically determining the language spoken in an audio signal based on its acoustic and phonetic characteristics, without relying on textual transcription or semantic understanding. Instead of recognizing words or sentences, LID systems focus on identifying language-specific patterns such as phoneme distributions, rhythmic structures, and intonation characteristics that distinguish one language from another.

LID plays a fundamental role in many modern speech-based applications. Voice assistants and smart devices use language identification to select appropriate speech recognition and synthesis models. Multilingual call centers rely on LID for automatic call routing, while speech analytics systems employ it to process large volumes of multilingual audio data efficiently. In such applications, accurate language identification is often a prerequisite for reliable downstream speech processing.

Controlled Speech Data vs. Real-World Audiobook Recordings

The performance and generalization capability of LID systems are strongly influenced by the characteristics of the speech data used during training and evaluation. Speech data recorded in controlled environments, such as studios or laboratories, typically exhibits high signal quality, minimal background noise, and limited variability in recording conditions. While these datasets simplify analysis and model development, they may not accurately represent the diversity and variability encountered in real-world scenarios.

In contrast, real-world speech data often includes variations in speaking style, prosody, and acoustic conditions. Audiobook recordings represent a semi-natural category of speech data: although professionally narrated and relatively clean, they preserve natural sentence-level structure, expressive intonation, and realistic linguistic patterns. As a result, audiobooks provide a practical compromise between fully controlled datasets and unconstrained real-world recordings, making them particularly suitable for language identification tasks that aim to balance realism and data quality.

Closed-set and Open-set Language Identification

Spoken Language Identification systems are commonly divided into closed-set and open-set paradigms based on their assumptions about the set of possible languages. In closed-set LID, the system assumes that every input speech sample belongs to one of a predefined set of known languages used during training. The goal is to classify each input into exactly one of these known language categories. This paradigm is well-suited for applications with a fixed and limited set of target languages.

Open-set LID extends this formulation by allowing the possibility that an input speech sample may belong to an unknown language that was not observed during training. In this case, the system must be capable of both identifying known languages and rejecting out-of-set samples. Open-set LID is conceptually more challenging, as it requires models to generalize beyond the training distribution and make reliable decisions under uncertainty.

Overview of Implementation Approaches

Implementation strategies for LID vary depending on whether a closed-set or open-set formulation is adopted. Closed-set LID is typically addressed using supervised classification approaches, where labeled speech data is used to train models that directly map extracted audio features to language labels. Common techniques include probabilistic models, margin-based classifiers, and neural network architectures.

Open-set LID often relies on similarity-based or distance-based approaches rather than direct classification. In these methods, speech samples are mapped into a feature or embedding space where language similarity can be measured. Decision mechanisms based on similarity scores or thresholds are then used to determine whether an input belongs to a known language or should be rejected as unknown. Such approaches provide greater flexibility in handling unseen languages and naturally connect to similarity learning techniques explored in later phases of the project.

Challenges in Language Identification

Phonetic Similarities Between Related Languages

One of the primary challenges in Spoken Language Identification arises from phonetic similarities between linguistically related languages. Languages that share common historical or geographical origins often exhibit overlapping phoneme inventories, similar syllable structures, and comparable prosodic patterns. These similarities can make it difficult for LID systems to distinguish between languages based solely on short acoustic segments, particularly when the speech content is limited or lacks distinctive phonetic cues.

In such cases, language-discriminative features may capture patterns that are shared across multiple languages rather than language-specific characteristics. This challenge is especially pronounced in short-duration recordings, where the statistical representation of phonetic patterns is inherently sparse. As a result, LID systems must rely on subtle differences in sound distribution, timing, and intonation, which may not always be reliably observable.

Accent and Speaker Variability

Another significant challenge in language identification is variability introduced by different speakers and accents. Speech signals are influenced not only by language but also by individual speaker characteristics such as pitch range, speaking rate, articulation style, and vocal tract shape. Additionally, accents and dialects within the same language can introduce acoustic variations that may resemble characteristics of other languages.

This variability can cause LID models to inadvertently learn speaker-specific or accent-specific patterns instead of language-dependent features. When training data does not adequately capture the diversity of speakers and accents, model performance may degrade when exposed to unseen speakers or speaking styles. Addressing this challenge requires careful dataset construction and feature selection to emphasize language-invariant yet language-discriminative properties of speech.

Short-Duration Speech Segments

Language identification performance is also strongly affected by the duration of available speech. Short audio segments provide limited information about phoneme distributions, rhythmic structure, and prosody, increasing uncertainty in language classification. In practical applications, LID systems often operate under time constraints, requiring reliable predictions from speech segments lasting only a few seconds or less.

With short-duration inputs, traditional statistical assumptions may no longer hold, and the extracted features may be less stable. This challenge motivates the need for robust feature representations that can capture discriminative language information even from brief speech samples. It also highlights the importance of consistent segmentation and careful selection of audio samples during dataset creation.

Research Directions and Solution Strategies

To address these challenges, existing research in language identification has explored a variety of solution strategies. One common approach is the use of robust acoustic features that capture both spectral and temporal characteristics of speech, reducing sensitivity to speaker and recording variability. Feature normalization and aggregation techniques are often employed to mitigate the effects of short-duration inputs.

Another prominent research direction focuses on learning compact and discriminative representations of speech through embedding-based methods. These approaches aim to map speech samples into a space where language-related similarities are emphasized while irrelevant variations are suppressed. Similarity-based learning frameworks further enable flexible decision-making, particularly in open-set LID scenarios.

Additionally, data-centric strategies such as careful dataset design, balanced language representation, and qualitative listening-based validation play a crucial role in improving LID performance. Together, these methods reflect a shift toward holistic language identification systems that combine informed data selection with robust modeling techniques.

Audio Data Characteristics and Selection

Importance of Proper Audio Selection for Language Identification

The effectiveness of Spoken Language Identification systems is fundamentally dependent on the quality and suitability of the audio data used during training and evaluation. Unlike many machine learning tasks where preprocessing techniques can partially compensate for noisy or suboptimal data, language identification relies on subtle acoustic and phonetic cues that may be irreversibly distorted by poor recording conditions or inappropriate segmentation. As a result, careful audio selection is a critical step that directly influences the discriminative power of extracted features and the reliability of subsequent modeling stages.

Improperly selected speech samples may introduce confounding factors such as excessive silence, background music, or abrupt segment boundaries that obscure language-specific characteristics. In contrast, well-chosen audio segments preserve natural linguistic patterns, allowing models to focus on meaningful differences between languages rather than artifacts introduced during data preparation. Therefore, dataset quality in LID is not solely determined by quantity, but by how well the audio content reflects authentic language use.

Desirable Properties of Speech Samples for LID Tasks

Speech samples intended for language identification should exhibit several key properties to ensure their effectiveness. First, the speech must be clear and intelligible, with minimal background noise or non-speech interference. This ensures that spectral and temporal features capture characteristics of the spoken language rather than recording artifacts. Second, samples should respect natural sentence boundaries, beginning at the start of a sentence and ending after its completion. Preserving sentence-level structure helps maintain natural prosody, rhythm, and intonation patterns that are informative for language discrimination.

Consistency in segment duration is also important, as it reduces variability introduced by differing amounts of speech content. Additionally, the selected audio should be representative of the target language, reflecting typical pronunciation patterns and speaking styles. In the context of audiobook recordings, this involves selecting narration segments that emphasize continuous speech rather than dramatic pauses or expressive non-linguistic elements. Together, these properties contribute to the creation of a balanced and informative dataset suitable for LID tasks.

Listening-Based Qualitative Evaluation

While technical criteria such as signal-to-noise ratio and segment duration are important, they are not sufficient on their own to guarantee dataset quality. Listening-based qualitative evaluation plays a complementary role by allowing human judgment to identify issues that may not be immediately detectable through automated analysis. By actively listening to each selected audio segment, it becomes possible to assess clarity, naturalness, and linguistic representativeness more reliably.

Qualitative evaluation enables the detection of subtle problems such as truncated words, unnatural pauses, or background sounds that may negatively impact feature extraction. This human-in-the-loop approach ensures that the collected dataset aligns with the intended objectives of language identification rather than relying solely on post-hoc signal processing. Incorporating listening-based validation during dataset construction strengthens the foundation of the entire machine learning pipeline and improves the robustness of subsequent modeling phases.

Feature Extraction Techniques

Feature extraction is a critical step in speech-based language identification, as raw audio waveforms are high-dimensional and not directly suitable for machine learning models. Effective features aim to capture language-dependent acoustic patterns while suppressing irrelevant variations such as speaker identity or recording conditions. This section reviews commonly used feature extraction techniques in LID, explaining their principles, suitability, advantages, and limitations.

Mel-Frequency Cepstral Coefficients (MFCC)

Mel-Frequency Cepstral Coefficients are among the most widely used features in speech and audio processing. MFCCs are computed by applying a short-time Fourier transform to the signal, mapping the power spectrum onto the Mel scale to approximate human auditory perception, and then applying a discrete cosine transform to decorrelate the resulting coefficients.

MFCCs are well-suited for language identification because they capture spectral envelope information related to phonetic structure, which varies across languages. They are effective in representing vowel and consonant characteristics that reflect language-specific phoneme distributions. However, MFCCs can be sensitive to noise and channel variations, and they may capture speaker-related information alongside language cues if not properly normalized.

Fast Fourier Transform (FFT)

The Fast Fourier Transform converts a time-domain signal into its frequency-domain representation, revealing the distribution of energy across frequencies. In speech processing, FFT is typically applied over short frames to analyze local spectral characteristics.

While FFT provides a fundamental representation of spectral content, it is generally not sufficient on its own for language identification. Raw frequency spectra lack perceptual scaling and temporal abstraction, making them sensitive to speaker variability and noise. FFT is more commonly used as a building block for higher-level features rather than as a standalone representation for LID.

Log-Mel Spectrogram

Log-Mel spectrograms represent the time-frequency energy distribution of speech after mapping frequencies to the Mel scale and applying logarithmic compression. This representation preserves both spectral and temporal information and is widely used in deep learning-based speech models.

Log-Mel spectrograms are highly suitable for language identification, particularly when used with convolutional or neural network architectures. They capture language-dependent spectral patterns while maintaining temporal context. However, they are high-dimensional and may require large amounts of data and careful regularization to avoid overfitting.

Spectral Centroid

The spectral centroid measures the center of mass of the spectrum and provides an estimate of where the dominant frequency components are concentrated. It is often interpreted as a measure of spectral brightness.

In the context of language identification, spectral centroid can capture coarse differences in articulation and phoneme distribution across languages. However, it is a low-level feature that lacks detailed phonetic information. As a result, it is typically insufficient on its own and is better used in combination with other spectral features.

Chroma Features

Chroma features represent the distribution of spectral energy across pitch classes and are commonly used in music analysis. They emphasize harmonic structure while reducing sensitivity to octave differences.

For language identification, chroma features are generally of limited usefulness, as speech signals are not primarily structured around harmonic pitch classes. While they may capture certain intonational patterns, chroma features do not directly encode phonetic or linguistic information and are therefore less effective for LID tasks.

Spectral Contrast

Spectral contrast measures the difference between peaks and valleys in the spectrum across frequency subbands. It captures relative spectral variations rather than absolute energy levels.

Spectral contrast can provide complementary information to traditional spectral features by highlighting structural differences in speech sounds. It may help distinguish languages with differing articulation styles or phoneme emphasis. However, its effectiveness depends on recording quality, and it is often used as an auxiliary feature rather than a primary representation.

Zero-Crossing Rate

The zero-crossing rate measures how frequently a signal changes sign within a given time frame. It provides a simple indication of signal noisiness and high-frequency content.

While zero-crossing rate is computationally inexpensive, it offers limited discriminatory power for language identification. It may capture differences between voiced and unvoiced speech segments but does not encode detailed linguistic information. Consequently, it is typically insufficient for LID when used in isolation.

Linear Predictive Coding (LPC)

Linear Predictive Coding models the speech signal as the output of a linear filter driven by an excitation source. LPC coefficients approximate the vocal tract characteristics of the speaker.

LPC can capture formant structure, which is relevant to phonetic content and thus to language identification. However, LPC features are sensitive to noise and speaker variability, and they may overemphasize speaker-specific characteristics if not carefully processed.

Perceptual Linear Prediction (PLP)

Perceptual Linear Prediction extends LPC by incorporating perceptual models of human hearing, such as critical-band analysis and equal-loudness preemphasis. The goal is to produce features that better align with auditory perception.

PLP features are well-suited for language identification, as they balance phonetic representation with perceptual relevance. Compared to LPC, PLP is more robust to noise and speaker variation. However, PLP computation is more complex, and its effectiveness depends on proper parameter selection and preprocessing.

Similarity Learning

Definition of Similarity Learning in Speech and Language Analysis

Similarity learning refers to a class of machine learning approaches in which the goal is not direct classification, but learning a representation space where similar inputs are mapped close to each other and dissimilar inputs are mapped far apart. In speech and language analysis, similarity learning aims to capture intrinsic relationships between speech samples by emphasizing task-relevant characteristics while suppressing irrelevant variability such as speaker identity or recording conditions.

Rather than assigning a fixed class label, similarity-based models learn embeddings that encode meaningful acoustic and linguistic properties. In the context of spoken language identification, this allows speech samples from the same language to cluster together in the embedding space, while samples from different languages become separable based on distance or similarity measures. This formulation is particularly well-suited to scenarios where flexibility and generalization are required.

Contrastive Loss and Triplet Loss

Similarity learning commonly relies on specialized loss functions designed to shape the embedding space. One widely used objective is contrastive loss, which operates on pairs of samples. Given a pair labeled as similar or dissimilar, the loss encourages similar pairs to have small distances in the embedding space while enforcing a minimum separation between dissimilar pairs. This pairwise formulation allows the model to learn explicit notions of similarity and dissimilarity.

Another popular objective is triplet loss, which operates on triplets of samples consisting of an anchor, a positive example, and a negative example. The anchor and positive samples belong to the same class, while the negative sample belongs to a different class. Triplet loss encourages the distance between the anchor and positive samples to be smaller than the distance between the anchor and negative samples by a predefined margin. This relative comparison framework often leads to more stable and discriminative embeddings, particularly in multi-class problems.

Application of Similarity Learning to Language Discrimination

Similarity learning provides a natural framework for language discrimination, especially in open-set language identification tasks. By learning a language-aware embedding space, the system can compare incoming speech samples against known language representations using distance-based criteria. This enables not only the identification of known languages but also the rejection of samples that do not sufficiently match any learned language cluster.

In closed-set scenarios, similarity-based embeddings can still be beneficial by improving class separability and robustness to speaker variability. In open-set settings, similarity learning enables flexible decision boundaries that are not restricted to predefined class labels. As a result, similarity learning serves as a powerful bridge between feature extraction and decision-making, supporting scalable and adaptable language identification systems in real-world applications.